

- Q.30 Describe the construction of a wing par.
 Q.31 How rigging of controls is done?
 Q.32 Name the components of mechanical control systems.
 Q.33 Explain the various types of minor defects and how they are reported.
 Q.34 What do you mean by periodical inspection?
 Q.35 Describe the construction of windows of aircrafts.

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 What are the types of undercarriages and their arrangements? Describe the construction and working of an olea-strut type landing gear.
 Q.37 Explain the use of non-metal material used in aircraft construction.
 Q.38 Explain the different types of wings, their attachment and construction details.

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4th Semester/ Aircraft Maintenance Subject : General Air Frame & Aero Moddling

Time : 3 Hrs.

M.M. : 100

SECTION-A

- Note:** Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 What material is used for aircraft fuselage?
 a) Aluminum alloys b) Titanium alloys
 c) Silver alloys d) Metal alloys
- Q.2 Stabilizing tail is also known as _____.
 a) Rudder b) Empennage
 c) Aileron d) Wingtip
- Q.3 Which of the following is not a retraction system for landing gear?
 a) Hydraulic retraction system
 b) Thermal retraction system
 c) Manual retraction system
 d) Electrical retraction system
- Q.4 Which was the first commercial aircraft with 50% of its structure weight made of carbon-fiber composite?
 a) Boeing 777 b) Boeing 787
 c) Boeing 747 d) Airbus A380

- Q.5 Which of the following is not true about an airframe made with carbon-fiber composite?
- Decreases drag
 - Decreases thrust
 - Higher cabin pressurization
 - Higher wing aspect ratio
- Q.6 The fuselage of an aircraft contains the engines.
- True
 - False
- Q.7 The earliest aircraft were constructed primarily of _____.
- Steel
 - Aluminum
 - Wood
 - None
- Q.8 What do winglets do?
- Increase thrust
 - Reduce turbulence at the tips of an airplane's wings
 - Decrease thrust
 - Increase turbulence at the tips of an airplane's wings
- Q.9 purpose of camber in an airfoil is _____.
- To increase maximum drag
 - To increase maximum lift
 - To decrease maximum lift
 - To decrease maximum drag
- Q.10 The term canard is used to describe the _____ configuration of an aircraft.
- Seating
 - Wing
 - Flap
 - Engine

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SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (10x1=10)

- Q.11 What is elevon?
- Q.12 Name two secondary controls in airplanes.
- Q.13 Plywood is used in _____ aircraft.
- Q.14 The purpose of tail rotor in helicopter is _____.
- Q.15 The tyres of airplanes are filled with _____.
- Q.16 Semi monocoque has _____ and _____.
- Q.17 Fuel stored in one/multiple tanks.
- Q.18 The main load bearing component of fuselage is _____?
- Q.19 Composites are lighter / heavier than metals.
- Q.20 Glider takes off with the help of _____.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Sketch and label various components of an aircraft.
- Q.22 Write a short note on 'Secondary control surfaces'.
- Q.23 Distinguish between fixed and retractable under carriage.
- Q.24 What do you mean by steel tubular construction.
- Q.25 Explain the various types of minor defects and how they are reported.
- Q.26 What do you mean by balancing of control surfaces?
- Q.27 What are the different types of fuel tanks?
- Q.28 How does a lighter than air aircraft fly.
- Q.29 Name the different composite material parts of aircrafts.

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